- *Roles* --The roles of DCC are defined by students, advisers, and anybody who wants to know more about the DCC certificate program. These people are associated with the university and identified in the student management system with a person number. In particular, it is expected that this tool is meant for and will be used by students interested or enrolled in the DCC program. The person-number identity of the university system is the tether linking the DCC application to the larger university management software. Each user of DCC holds a decentralized (self-generated) globally unique identity of 256 bits, and this identity is mapped on their local university identity of the person number. (Recall the generation of a decentralized identity in chapter 8 using a web tool.)
- *Rules* --The first rule is about the identity of the user. A student self-registers with the person number, an identity assigned by a university system. This rule enables the decentralized blockchain identity of an enrollee to be mapped onto the person number of the centralized system. The second rule is that only a registered user (`validStudent`) can add courses to the DCC application.

  NOTE The `msg.sender` attribute of a blockchain transaction holds the 256-bit decentralized identity of the sender of the transaction.

- *Assets* --Assets are the course categories, allowed courses in each category, and GPA. Each student user with a (blockchain-based) decentralized identity has their own set of data:
- An efficient structure of courses and grades
- A mapping of their 256-bit address to corresponding person number